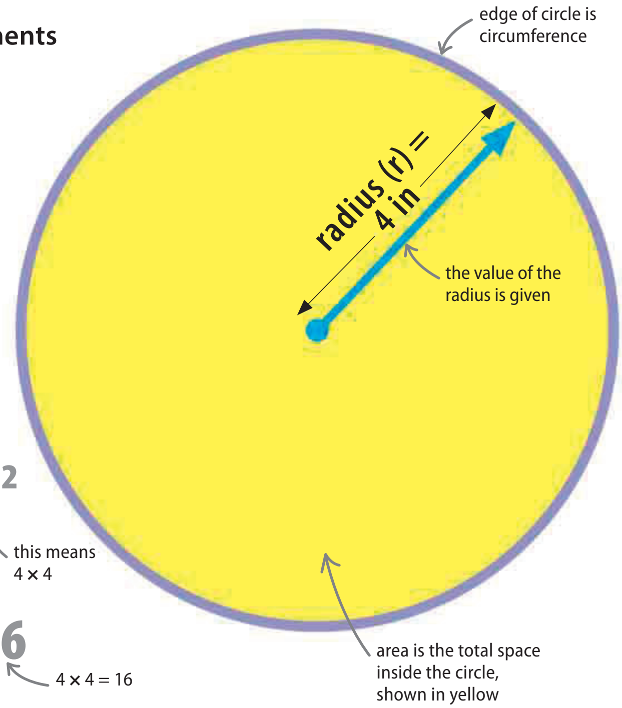
Area of a circle

THE AREA OF A CIRCLE IS THE AMOUNT OF SPACE ENCLOSED INSIDE ITS PERIMETER (CIRCUMFERENCE).

SEE ALSO
❮ 138–139 Circles
❮ 140–141 Circumference and diameter
Formulas 177–179 ❯

The area of a circle can be found by using the measurements of either the radius or the diameter of the circle.



Finding the area of a circle

The area of a circle is measured in square units. It can be found using the radius of a circle (r) and the formula shown below. If the diameter is known but the radius is not, the radius can be found by dividing the diameter by 2.

In the formula for the area of a circle, πr^2 means π (pi) × radius × radius.

area of a circle; π is a fixed value; radius

$$\text{area} = \pi r^2$$

Substitute the known values into the formula; in this example, the radius is 4 in.

$$\text{area} = 3.14 \times 4^2$$

π is 3.14 to 3 significant figures; a more accurate value can be found on a calculator

this means 4×4

Multiply the radius by itself as shown—this makes the last multiplication simpler.

$$\text{area} = 3.14 \times 16$$

$4 \times 4 = 16$

Make sure the answer is in the right units (in^2 here) and round it to a suitable number.

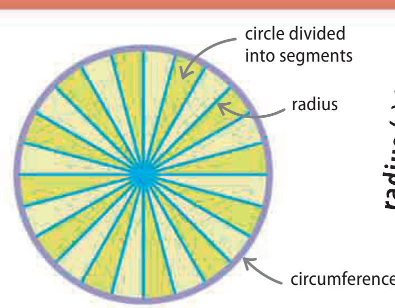
$$\text{area} = 50.24\ \text{in}^2$$

answer is 50.24 exactly

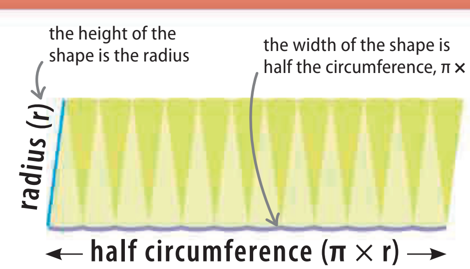
LOOKING CLOSER

Why does the formula for the area of a circle work?

The formula for the area of a circle can be proved by dividing a circle into segments, and rearranging the segments into a rectangular shape. The formula for the area of a rectangle is simpler than that of the area for a circle—it is just height × width. The rectangular shape's height is simply the length of a circle segment, which is the same as the radius of the circle.
The width of the rectangular shape is half of the total segments, equivalent to half the circumference of the circle.



Split any circle up into equal segments, making them as small as possible.



Lay the segments out in a rectangular shape. The area of a rectangle is height × width, which in this case is radius × half circumference, or $r \times \pi r$, which is πr^2.